

SOFTWARE REQUIREMENTS DOCUMENT

AI-Powered Student Study Assistant

"AI Buddy"

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Field	Details
Project	AI Buddy — AI-Powered Student Study Assistant
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Departments	MBA · BBA · B.Tech · MCA · Hotel Management
Student Scope	3,000+ students
LLM Stack	Google Gemini 2.0 Flash (Primary) · Groq / Llama 3.3 70B (Fallback)
Vector DB	MongoDB Atlas Vector Search (M0 Free Tier)
Embedding Model	Google text-embedding-004
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1. Introduction & Purpose

This Software Requirements Document (SRD v2.0) translates the AI Buddy BRD v2.0 into developer-ready specifications. It covers every screen, service, schema, API endpoint, and integration the engineering team must build to deliver AI Buddy to 3,000+ students across five departments.

⚠ Note: The BRD specifies 'LLM' generically. This SRD formally selects Google Gemini 2.0 Flash as the primary free LLM and Groq (Llama 3.3 70B) as fallback. No proprietary paid LLM is used — both are free-tier in test and production.

1.1 Scope of This Document

- Complete folder structure (monorepo) with purpose of every file
- 28 application pages — route, role, components, behaviour
- All 13 MongoDB collection schemas with field-level types and constraints
- LLM/RAG pipeline: ingestion → embedding → retrieval → Gemini MCQ generation
- Nodemailer SMTP mailing system — 8 email trigger events
- Community forum, weekly feedback form, and admin feedback viewer
- Exam lockout logic, ethics detection, safety module, and full RBAC matrix

1.2 Technology Stack (Locked)

Layer	Technology	Version / Detail	Cost
Frontend	React.js	v18+	Free / OSS
Backend	Node.js + Express	LTS	Free / OSS
Primary LLM	Google Gemini 2.0 Flash	Gemini API	Free tier — 1,500 req/day, 1M ctx
Fallback LLM	Groq — Llama 3.3 70B	Groq API	Free tier — ~14,400 req/day
Embedding Model	Google text-embedding-004	Gemini API	Free with Gemini API key
Vector DB	MongoDB Atlas Vector Search	M0 cluster	Free forever on M0 tier
Primary DB	MongoDB Atlas	v7+	Free M0 (512MB)
Auth	Institutional SSO	OAuth 2.0	Institution-provided
Mail	Nodemailer + Local SMTP	Latest	Free — institution SMTP
Cloud	AWS / Azure (TBD)	—	Institution budget
Security	TLS 1.2+ / AES-256	—	Infrastructure layer

2. System Overview & Architecture

2.1 Three-Tier Architecture

Tier	Technology	Responsibility
Client (Web/Mobile)	React.js v18+	Student portal, admin dashboard, forum, feedback viewer, safety panel
API Server	Node.js + Express (LTS)	REST API, LLM orchestration, SMTP mailer, RBAC middleware, cron jobs
AI Layer	Gemini 2.0 Flash + RAG	MCQ generation, learning paths, AI tutor Q&A, distress classification
Data Layer	MongoDB Atlas (Vector + Document)	User profiles, question bank, audit logs, forum, feedback, schemas
Auth Layer	Institutional SSO (OAuth 2.0)	Login, JWT issuance, role mapping (student/faculty/HOD/admin)
Mail Layer	Nodemailer + Local SMTP	All transactional emails — sessions, escalations, feedback reminders

2.2 Request Lifecycle

Step	Actor	Action
1	Student	Authenticates via Institutional SSO → JWT issued (role + dept + semester + subjects)
2	React Client	Sends API request with Bearer JWT to Express server
3	RBAC Middleware	Validates JWT, checks role permissions, checks logout status
4	Controller	Routes request to appropriate service (LLM / DB / Mail)
5	LLM Service	Embeds query → vector search → assembles RAG context → calls Gemini API
6	Gemini 2.0 Flash	Generates MCQ / learning path / tutor response grounded in RAG context
7	Audit Service	Writes immutable log entry (timestamp + source refs + response)
8	Response	Result returned to React client; ethics/distress checks fire asynchronously

2.3 AI Architecture — RAG Pipeline Overview

Primary LLM: Google Gemini 2.0 Flash — free, 1M token context, native JSON mode

Fallback LLM: Groq Llama 3.3 70B — free, ~200 tok/sec, OpenAI-compatible API

Embeddings: Google text-embedding-004 — free, 768 dimensions, same key as Gemini

Vector Search: MongoDB Atlas Vector Search on M0 free cluster — no extra infra

The RAG pipeline keeps ALL AI outputs grounded in institution-provided content (syllabi, notes, past papers). This directly addresses the BRD's hallucination-accountability SLA requirement.

3. Folder Structure — Complete Map

Every directory and file is listed below with its purpose. Blue = directory, monospace = file name. Use this as the definitive reference for sprint task assignments.

3.1 Root Monorepo

Path	Purpose
 ai-buddy/	Root monorepo
├─ client/	React.js frontend application
├─ server/	Node.js + Express backend API
├─ shared/	Shared types, constants, Zod validation schemas
├─ scripts/	DB seed scripts, data ingestion pipelines
├─ docs/	SRD, BRD, API specs, architecture diagrams
├─ .env.example	Environment variable template (never commit .env)
├─ .gitignore	
├─ package.json	Monorepo root — npm workspaces config
└─ README.md	Project setup and onboarding guide

3.2 Backend — server/

3.2.1 Config Layer — server/src/config/

Path	Purpose
 server/src/config/	Application configuration files
├─ db.js	MongoDB Atlas connection via Mongoose; handles retry logic
├─ mailer.js	Nodemailer SMTP transporter setup; TLS 1.2+ enforced
├─ llm.js	Gemini API + Groq API client init; fallback switcher logic
├─ vectorDb.js	MongoDB Atlas Vector Search index config; embedding dimensions
└─ env.js	Env var loader + Zod validation; app crashes if required vars missing

3.2.2 Models Layer — server/src/models/

Path	Purpose
 server/src/models/	Mongoose schema definitions — see Section 6 for full schemas
├─ User.js	Students, Faculty, HODs, Admins — unified with role discriminator
├─ Session.js	Auth sessions + JWT metadata + refresh tokens
├─ QuestionBank.js	10-year past paper questions — tagged by dept/subject/topic/year
├─ KnowledgeChunk.js	RAG content chunks — text + vector embedding + metadata
├─ LearningPath.js	AI-generated personalised revision paths per student
├─ MockTest.js	Generated MCQ test instances — questions + student answers + score

Path	Purpose
└─ AuditLog.js	Immutable log of every AI interaction — never deleted
└─ EthicsFlag.js	Flagged unethical request records + HOD escalation status
└─ EthicsConfig.js	Per-dept prohibited request category definitions (versioned)
└─ DoubtSession.js	Faculty doubt-clearing session bookings + meeting links
└─ FacultyAvailability.js	Faculty availability windows per day/week
└─ ForumPost.js	Community forum posts, replies, grievance flags
└─ Grievance.js	Formal grievance tickets with reference numbers + status
└─ FeedbackForm.js	Weekly student feedback submissions
└─ EmergencyContact.js	Hospital / police / fire contact data per campus
└─ ExamTimetable.js	Per-student, per-subject exam schedule (lockout source of truth)
└─ HallucinationReport.js	Student-submitted AI hallucination reports

3.2.3 Routes Layer — server/src/routes/

Path	Purpose
📁 server/src/routes/	Express router files — one per domain
└─ auth.routes.js	POST /api/auth/login /logout /refresh
└─ student.routes.js	Student learning path, question bank, mock test, sessions
└─ faculty.routes.js	Faculty availability management and session handling
└─ hod.routes.js	HOD audit log, ethics flag review, ethics config
└─ admin.routes.js	System dashboard, user mgmt, timetable upload, emergency contacts
└─ forum.routes.js	Forum CRUD — posts, replies, grievance escalation
└─ feedback.routes.js	POST /api/forum/feedback GET admin feedback viewer
└─ llm.routes.js	MCQ generation, mock test, learning path, AI tutor chat
└─ safety.routes.js	Distress detection confirm + emergency contact fetch
└─ lockout.routes.js	Exam lockout status check + admin timetable sync
└─ hallucination.routes.js	Student hallucination report submission

3.2.4 Controllers Layer — server/src/controllers/

Path	Purpose
📁 server/src/controllers/	Business logic — mirrors routes/ one-to-one
└─ auth.controller.js	SSO callback, JWT generation, session management
└─ student.controller.js	Learning path fetch, question bank query, mock test flow
└─ faculty.controller.js	Availability CRUD, session confirm/decline, meeting link
└─ hod.controller.js	Audit log queries, flag review, ethics config updates
└─ admin.controller.js	Dashboard aggregates, user CRUD, timetable upload, contacts
└─ forum.controller.js	Post CRUD, reply threading, grievance creation
└─ feedback.controller.js	Feedback submit, admin viewer query, CSV export
└─ llm.controller.js	Orchestrates RAG → Gemini pipeline per request type
└─ safety.controller.js	Distress confirmation flow, emergency contact fetch
└─ lockout.controller.js	Per-student lockout status check
└─ hallucination.controller.js	Report save + vendor/IT email trigger

3.2.5 Services Layer — server/src/services/

Path	Purpose
📁 server/src/services/	Pure business logic called by controllers
├─ llm.service.js	Gemini API calls; Groq fallback on 429/5xx; retry with backoff
├─ rag.service.js	Embed query → Atlas Vector Search → assemble context block
├─ mcq.service.js	MCQ generation pipeline: RAG context + Gemini + schema validation
├─ embedding.service.js	text-embedding-004 calls; batch upsert to Atlas Vector Search
├─ mailer.service.js	8 email templates + send functions (see Section 7)
├─ lockout.service.js	Read ExamTimetable → compute active lockout windows per student
├─ ethics.service.js	Pattern match + Gemini classifier → flag + queue HOD escalation
├─ safety.service.js	Distress signal detection + two-step escalation flow
├─ audit.service.js	Immutable AuditLog writer — called synchronously before response
├─ meeting.service.js	Google Meet / Zoom / Teams API call → return meeting link
└─ forum.service.js	Forum logic, grievance routing, feedback weekly trigger

3.2.6 Middleware Layer — server/src/middleware/

Path	Purpose
📁 server/src/middleware/	Express middleware — applied per route group
├─ auth.middleware.js	Verify JWT; attach req.user (id, role, dept, semester, subjects)
├─ rbac.middleware.js	Role-based access check; returns 403 on violation
├─ lockout.middleware.js	Blocks /api/llm/* and AI routes during student exam window
├─ deptScope.middleware.js	Enforces dept filter on all queries — prevents cross-dept access
└─ rateLimiter.js	express-rate-limit — 100 req/15min per IP; stricter on /api/llm/*

3.2.7 Cron Jobs — server/src/jobs/

Path	Purpose
📁 server/src/jobs/	Scheduled background tasks using node-cron
├─ lockoutCron.js	Runs every 15 min — reads ExamTimetable; sets/clears lockout flags
└─ feedbackCron.js	Runs every Monday 9:00 AM IST — sets feedback_due=true for active students

3.2.8 Root Files — server/

Path	Purpose
📁 server/	
├─ app.js	Express app init — mounts all routers, middleware, error handler
├─ server.js	HTTP server entry point — starts app + cron jobs
├─ tests/	Jest unit + integration tests mirroring services/ structure
└─ package.json	Backend dependencies (express, mongoose, nodemailer, @google/generative-ai, groq-sdk, bull, node-cron)

3.3 Frontend — client/

3.3.1 Pages — client/src/pages/

Path	Purpose
client/src/pages/	One file per screen — see Section 4 for full detail
└─ auth/	login.jsx unauthorized.jsx
└─ student/	dashboard, learning-path, question-bank, practice-mcq, mock-test, mock-test-results, ai-tutor, videos, book-session, my-sessions, forum, report-hallucination, emergency (13 pages)
└─ faculty/	dashboard, availability, session-requests, my-sessions (4 pages)
└─ hod/	dashboard, audit-log, ethics-config, faculty-mgmt (4 pages)
└─ admin/	dashboard, users, timetable, emergency-contacts, feedback (5 pages)

3.3.2 Components — client/src/components/

Path	Purpose
client/src/components/	Reusable UI components
└─ layout/	Navbar.jsx Sidebar.jsx Footer.jsx PageWrapper.jsx
└─ ui/	Button Modal Card Badge Toast Spinner Tooltip
└─ mcq/	MCQCard.jsx OptionButton.jsx TimerBar.jsx ResultCard.jsx
└─ forum/	PostCard.jsx ReplyThread.jsx GrievanceTag.jsx NewPostForm.jsx
└─ feedback/	WeeklyFeedbackModal.jsx RatingInput.jsx FeedbackTable.jsx
└─ safety/	EmergencyPanel.jsx DistressPrompt.jsx ContactCard.jsx

3.3.3 Other Frontend Dirs

Path	Purpose
client/src/	
└─ hooks/	useAuth.js useLockout.js useForum.js useDebounce.js
└─ store/	Zustand stores: authStore.js learningStore.js forumStore.js
└─ api/	Axios client functions mirroring all backend route groups
└─ utils/	dateHelpers.js formatters.js validators.js constants.js
└─ styles/	global.css tailwind.config.js theme tokens

3.4 Shared & Scripts

Path	Purpose
shared/	Used by both client and server
└─ schemas/	Zod validation schemas: mcqSchema.js feedbackSchema.js userSchema.js
└─ constants/	ROLES, DEPARTMENTS, SEMESTERS, ERROR_CODES, EMAIL_TRIGGERS
└─ types/	JSDoc type definitions for User, MCQ, LearningPath, AuditLog
scripts/	Run locally or in CI — not deployed to production
└─ seedDb.js	Seed dev DB with mock students, faculty, question bank samples

Path	Purpose
 ingestContent.js	<i>Reads PDFs/DOCX from /data — chunks, embeds, upserts to Atlas</i>
 validateEnv.js	<i>Pre-start check: confirms all required .env vars are present</i>

4. Application Pages — Screen Inventory (28 Pages)

Total screens: 28 across 5 role groups. Each entry specifies the route path, allowed role(s), primary purpose, and key UI components that must be built.

4.1 Authentication (2 pages)

#	Page	Route	Roles	Purpose & Key Components
1	Login	/login	All	SSO OAuth 2.0 redirect. Institutional login button, redirects post-auth by role (student→/student/dashboard, faculty→/faculty/dashboard, etc.)
2	Unauthorized	/unauthorized	All	Shown on forbidden route access. Error message, back-to-dashboard link, contact IT prompt.

4.2 Student Pages (13 pages)

#	Page	Route	Purpose & Key Components
3	Dashboard	/student/dashboard	Home screen: AI-generated today's study agenda, exam countdown timers per subject, lockout status banner (if active), floating Emergency button (always visible).
4	Learning Path	/student/learning-path	AI-generated personalised revision roadmap. Ordered topic list (weakest first), progress bar per topic, estimated time-to-completion, 'Start Practice' shortcut per topic.
5	Question Bank	/student/question-bank	Browse 10-year past paper bank. Filters: Department / Subject / Topic / Year / Difficulty. Card grid view per question; expand to full text. Bookmark questions.
6	Practice MCQ	/student/practice/mcq	Topic-wise MCQ session. Single question per view, 4-option radio buttons, Next / Submit controls, instant feedback after each answer with explanation + source citation.
7	Mock Test	/student/mock-test	Full timed mock exam (30–50 Qs). Config screen: subject/duration select. Countdown timer (TimerBar), auto-submit on timeout, no back-navigation mid-test.
8	Mock Test Results	/student/mock-test/results/:id	Post-test analytics. Score donut chart, correct/incorrect breakdown by topic, weak area highlight cards, PDF download, 'Practice Weak Topics' CTA.
9	AI Tutor Chat	/student/ai-tutor	RAG-powered concept Q&A chat. Message thread UI, streaming response, source citation chip on every AI message, 'Report Hallucination' icon per response.

#	Page	Route	Purpose & Key Components
10	Video Resources	/student/videos	Curated licensed video links per subject. Card grid with thumbnail preview, title, duration label. Opens in new tab (no embedded player).
11	Book Session	/student/book-session	Faculty doubt-clearing booking. Faculty list (dept-filtered), availability calendar, subject/doubt topic text input, Submit button triggers E-01 email.
12	My Sessions	/student/sessions	History of booked sessions. Table: Faculty Subject Date Status (pending/confirmed/completed) Meeting Link (active when confirmed).
13	Forum	/student/forum	Dept-scoped peer Q&A. Tab: All Posts / My Posts. New Post button, reply threads, Grievance Flag button per post. Weekly feedback modal auto-shown on Mondays.
14	Report Hallucination	/student/report-hallucination	Hallucination report form. Read-only AI response text, student description textarea, screenshot upload (optional), Submit triggers vendor+IT email + AuditLog entry.
15	Emergency & Safety	/student/emergency	24/7 emergency panel (never locked out). Hospital / police / fire contact cards with tap-to-call and Maps directions link. Distress check-in section with 2-step confirm flow.

4.3 Faculty Pages (4 pages)

#	Page	Route	Purpose & Key Components
16	Dashboard	/faculty/dashboard	Overview: pending session request count badge, today's confirmed sessions list, availability status toggle (Available / Unavailable today).
17	Set Availability	/faculty/availability	Weekly calendar grid. Click slots to toggle available/unavailable. Save button → updates FacultyAvailability collection. Changes reflected in student booking flow immediately.
18	Session Requests	/faculty/session-requests	Incoming doubt requests table: Student ID Subject Doubt Topic Requested Time Actions (Confirm / Decline). Confirm triggers meeting link generation + E-02 emails.
19	My Sessions	/faculty/sessions	All sessions history: confirmed + completed + declined. Meeting link access for confirmed sessions. Notes field per session for post-meeting follow-up.

4.4 HOD Pages (4 pages)

#	Page	Route	Purpose & Key Components
20	Dashboard	/hod/dashboard	Dept KPIs: active students this week, total sessions booked, ethics flags (last 30 days), pending escalations badge. Quick links to audit log and ethics config.
21	Ethics & Audit Log	/hod/audit-log	Full immutable interaction log. Columns: Timestamp Student ID Request Summary Flag Category Status. Filters by date range, category, status. Export CSV. Mark false positive.
22	Ethics Config	/hod/ethics-config	Define prohibited request categories for this department. Add / edit / disable category rules. Each rule has: name, description, keywords list, severity (warn / block). Version history shown.
23	Faculty Management	/hod/faculty	Dept faculty list: name, availability status, sessions this month, fulfillment rate %. Read-only view to monitor faculty load and session completion rates.

4.5 Admin / IT Pages (5 pages)

#	Page	Route	Purpose & Key Components
24	Admin Dashboard	/admin/dashboard	System-wide KPIs: total active sessions, engagement rate %, uptime %, open ethics flags, hallucination reports, average feedback rating. Charts for weekly trends.
25	User Management	/admin/users	Full CRUD for all users across all depts. Table with role/dept/semester filters. Inline role edit, bulk CSV import, account enable/disable, password reset trigger.
26	Exam Timetable	/admin/timetable	Upload exam schedule CSV (format: student_id, subject, exam_date, start_time, end_time). Preview computed lockout windows before publishing. Manual per-student unlock with audit trail.
27	Emergency Contacts	/admin/emergency-contacts	CRUD for hospital / police / fire contacts per campus. Fields: name, phone, address, maps_link, contact_type. Live preview of how Emergency Panel will display to students.
28	Feedback Viewer	/admin/feedback	All weekly feedback submissions. Filterable by week / dept / rating. Average rating cards at top. Full table below: student_id (admin only) week ratings comments. CSV export.

Page Count: Auth: 2 · Student: 13 · Faculty: 4 · HOD: 4 · Admin: 5 · TOTAL: 28 screens

5. LLM / RAG Integration — Gemini + Groq

All AI outputs are grounded in institution-approved content via RAG. The primary LLM is Google Gemini 2.0 Flash (free). Groq Llama 3.3 70B is the automatic fallback on rate-limit or API failure.

5.1 Content Ingestion Pipeline

Step	Action	Tool / Service	Output
1	Collect source documents	Academic Registrar → /data folder	PDFs, DOCX: syllabi, notes, textbooks, past papers
2	Parse and extract text	pdf-parse + mammoth (Node.js)	Raw text per document
3	Chunk into segments	scripts/ingestContent.js	~500 token chunks, 50 token overlap
4	Tag metadata	ingestContent.js	{ dept, subject, topic, year, doc_type, source_file }
5	Generate embeddings	Google text-embedding-004 (free)	768-dim float32 vector per chunk
6	Upsert to Vector DB	MongoDB Atlas Vector Search	KnowledgeChunk documents with vector field
7	Verify	scripts/validateIngestion.js	Chunk count per dept/subject logged to console

5.2 RAG Query Flow

Step	Code Location	Action
1	llm.controller.js	Receive student request: { query, dept, subjects[], topic? }
2	rag.service.js	Embed query using text-embedding-004
3	rag.service.js	Atlas Vector Search: filter { dept, subject } → top K=5 chunks
4	rag.service.js	Assemble context block: concatenate chunk texts + metadata citations
5	mcq.service.js / llm.service.js	Build final prompt: system prompt + context block + user query
6	llm.service.js	Call Gemini 2.0 Flash API with JSON mode enabled
7	llm.service.js	On 429 / 5xx: automatic fallback to Groq Llama 3.3 70B
8	audit.service.js	Write AuditLog entry synchronously before returning response
9	llm.controller.js	Return response + source_refs[] to client

5.3 Gemini API Configuration

```
// server/src/config/llm.js
const { GoogleGenerativeAI } = require('@google/generative-ai');
const Groq = require('groq-sdk');

const gemini = new GoogleGenerativeAI(process.env.GEMINI_API_KEY);
const groq = new Groq({ apiKey: process.env.GROQ_API_KEY });

// Primary: Gemini 2.0 Flash – free tier, 1M token context
const primaryModel = gemini.getGenerativeModel({
  model: 'gemini-2.0-flash',
  generationConfig: {
    responseMimeType: 'application/json', // native JSON mode
    temperature: 0.2, // low = deterministic MCQs
    maxOutputTokens: 1500,
  },
});

// Fallback: Groq Llama 3.3 70B – ~14,400 req/day free, OpenAI-compatible
const fallbackCall = async (prompt) =>
  groq.chat.completions.create({
    model: 'llama-3.3-70b-versatile',
    messages: [{ role: 'user', content: prompt }],
    temperature: 0.2,
    response_format: { type: 'json_object' },
  });
```

5.4 MCQ Generation System Prompt Template

```
You are an academic question generator for {department} students,
subject: {subject}, topic: {topic}.

RULES:
1. Use ONLY the context provided below – no outside knowledge.
2. Generate exactly {count} multiple-choice questions.
3. Each question must have: stem, 4 options (A-D), correct answer key,
   one-sentence explanation citing the source chunk ID.
4. Vary difficulty: 40% easy, 40% medium, 20% hard.
5. Output ONLY valid JSON – no markdown, no preamble.
6. Match the schema exactly – extra fields will cause a 400 error.

CONTEXT:
{rag_context}

OUTPUT SCHEMA (array of N objects):
[ { question_id, stem, options:{A,B,C,D}, correct, explanation,
  difficulty, rag_sources:[] } ]
```

5.5 MCQ JSON Schema (Single Question)

Field	Type	Example / Notes
question_id	String (UUID)	Auto-generated by server before DB insert
dept	String	'B.Tech' — from student profile
subject	String	'Data Structures'

Field	Type	Example / Notes
topic	String	'Binary Search Trees'
stem	String	The question text (max 300 chars)
options.A	String	Option A text
options.B	String	Option B text
options.C	String	Option C text
options.D	String	Option D text
correct	String (enum)	'A' 'B' 'C' 'D'
explanation	String	One sentence. Must cite source: '[Source: DS Notes Ch.4 p.87]'
difficulty	String (enum)	'easy' 'medium' 'hard'
generated_at	ISO 8601 DateTime	Server timestamp of generation
rag_sources	String[]	Array of KnowledgeChunk _id values used for this question

5.6 Learning Path Generation

Aspect	Detail
Trigger	On first login after enrollment; after each mock test completion
Input Persona	{ dept, semester, enrolled_subjects[], weak_topics[], mock_scores[], exam_schedule[] }
System Prompt Goal	Order topics by priority: weakest first, exam-date deadline-aware, estimated study hours per topic
Output	Ordered topic array with: topic name, priority score, recommended resources (chunk IDs), estimated hours, exam_date deadline
Storage	LearningPath collection — one document per student, replaced on regeneration
Re-generation	Triggered by: new mock test result, HOD syllabus update, student manual refresh (max 1/day)

5.7 Hallucination Handling

Mechanism	Implementation
Source citations	Every AI response includes rag_sources[] — chunk IDs shown as footnotes in UI
Student reporting	'Report Hallucination' button on every AI Tutor message → /api/hallucination/report
Audit trail	HallucinationReport saved + AuditLog entry flagged 'hallucination'

Mechanism	Implementation
Vendor notification	Nodemailer sends E-05 email to vendor contact + IT immediately on submit
SLA tracking	SM-006 metric: reports submitted vs resolved; tracked in admin dashboard
Fallback	If Gemini response fails JSON schema validation: reject, log error, return fallback message

6. MongoDB Schemas — All Collections

All 13 Mongoose schemas are defined below. Required fields are marked ✓. All collections have automatic createdAt / updatedAt via mongoose timestamps:true option. All _id fields are ObjectId (auto).

Convention: ObjectId refs use 'ref: CollectionName'. Enums listed are exhaustive. max: constraints enforced at Mongoose level and re-validated in controller.

6.1 User (students, faculty, HODs, admins)

Field	Type	Required	Description / Constraints
_id	ObjectId	✓	Auto-generated primary key
institution_id	String	✓	Unique institutional login ID (from SSO)
name	String	✓	Full display name
email	String	✓	Institutional email — unique index
role	String (enum)	✓	'student' 'faculty' 'hod' 'admin'
dept	String (enum)	✓	'MBA' 'BBA' 'BTech' 'MCA' 'HotelMgmt'
semester	Number	Cond.	1–8. Required if role === 'student'
enrolled_subjects	String[]	Cond.	Required if role === 'student'. Validated against dept syllabus
weak_topics	String[]	–	Maintained by LLM service; updated after each mock test
feedback_due	Boolean	–	Set true every Monday by feedbackCron; reset false on submission
is_active	Boolean	✓	Default true; set false for suspended/graduated accounts
last_login	Date	–	Updated on every successful auth
created_at	Date	✓	Auto via timestamps:true
updated_at	Date	✓	Auto via timestamps:true

6.2 KnowledgeChunk (RAG content store)

Field	Type	Required	Description / Constraints
_id	ObjectId	✓	Auto
dept	String (enum)	✓	Department scope of this chunk

Field	Type	Required	Description / Constraints
subject	String	✓	Subject name
topic	String	✓	Topic / chapter within subject
year	Number	–	For past paper chunks: exam year (e.g., 2019)
doc_type	String (enum)	✓	'notes' 'textbook' 'syllabus' 'past_paper'
source_file	String	✓	Original filename for attribution
page_number	Number	–	Page reference for citation
text	String	✓	Chunk text content (max ~500 tokens)
embedding	Number[]	✓	768-dim float32 vector from text-embedding-004
created_at	Date	✓	Auto

6.3 QuestionBank (10-year past papers)

Field	Type	Required	Description / Constraints
_id	ObjectId	✓	Auto
dept	String (enum)	✓	Department
subject	String	✓	Subject name
topic	String	✓	Topic tag
year	Number	✓	Exam year
exam_type	String (enum)	✓	'mid_sem' 'end_sem' 'internal'
stem	String	✓	Question text
options	Object	Cond.	{ A, B, C, D } — required if MCQ type
correct	String (enum)	Cond.	'A' 'B' 'C' 'D' — required if MCQ type
answer_key	String	Cond.	For non-MCQ questions: model answer text
marks	Number	✓	Marks allocated to this question
difficulty	String (enum)	–	'easy' 'medium' 'hard' — AI-tagged on ingestion
created_at	Date	✓	Auto

6.4 LearningPath (per-student AI revision plan)

Field	Type	Required	Description / Constraints
_id	ObjectId	✓	Auto
student_id	ObjectId (ref: User)	✓	Foreign key to student user
dept	String	✓	Denormalised for filtering
semester	Number	✓	Denormalised
topics	Array of Objects	✓	Ordered list — see sub-schema below
topics[].topic_name	String	✓	Topic name
topics[].subject	String	✓	Subject the topic belongs to
topics[].priority_score	Number	✓	0–100; higher = weaker / more urgent
topics[].estimated_hours	Number	✓	AI estimate in hours
topics[].exam_deadline	Date	–	Nearest exam date for this subject
topics[].status	String (enum)	✓	'not_started' 'in_progress' 'completed'
generated_at	Date	✓	Timestamp of last AI generation
trigger	String (enum)	✓	'first_login' 'mock_test_complete' 'manual_refresh'

6.5 MockTest (generated test instances)

Field	Type	Required	Description / Constraints
_id	ObjectId	✓	Auto
student_id	ObjectId (ref: User)	✓	Who took the test
dept	String	✓	Denormalised
subject	String	✓	Subject scope
questions	Array of Objects	✓	MCQ objects (same schema as Section 5.5 — embedded)
student_answers	Object	–	{ question_id: 'A' 'B' 'C' 'D' } — populated on submit
score	Number	–	Computed on submit: correct / total * 100
topic_breakdown	Array of Objects	–	[{ topic, correct, total }] — for results page
duration_minutes	Number	✓	Test duration configured by student
started_at	Date	✓	Server timestamp on test start

Field	Type	Required	Description / Constraints
submitted_at	Date	–	Set on submit or auto-submit
status	String (enum)	✓	'in_progress' 'submitted' 'timed_out'

6.6 AuditLog (immutable — never deleted)

Field	Type	Required	Description / Constraints
_id	ObjectId	✓	Auto
student_id	ObjectId (ref: User)	✓	
session_id	String	✓	JWT session identifier
request_text	String	✓	Student's original query (max 2000 chars)
response_text	String	✓	AI response (truncated to 4000 chars if longer)
rag_sources	String[]	–	KnowledgeChunk_ids used in this response
flag_type	String (enum)	–	'ethics' 'hallucination' 'distress' null
is_flagged	Boolean	✓	Default false; set true on any flag_type
escalated_to	String	–	HOD email / security contact if escalated
escalated_at	Date	–	Timestamp of escalation dispatch
timestamp	Date	✓	Server timestamp — immutable after write

6.7 EthicsFlag (flagged unethical requests)

Field	Type	Required	Description / Constraints
_id	ObjectId	✓	Auto
audit_log_id	ObjectId (ref: AuditLog)	✓	Points to the full interaction record
student_id	ObjectId (ref: User)	✓	
dept	String	✓	For HOD filtering
category	String	✓	Matched prohibited category name (from EthicsConfig)
severity	String (enum)	✓	'warn' 'block'
hod_notified_at	Date	–	Timestamp when E-04 email was dispatched

Field	Type	Required	Description / Constraints
hod_review_status	String (enum)	✓	'pending' 'reviewed' 'false_positive' 'actioned'
hod_notes	String	–	HOD's review notes (max 1000 chars)
created_at	Date	✓	Auto

6.8 FacultyAvailability (faculty open slots)

Field	Type	Required	Description / Constraints
_id	ObjectId	✓	Auto
faculty_id	ObjectId (ref: User)	✓	
dept	String	✓	Denormalised
slots	Array of Objects	✓	Available time slots
slots[].day	String (enum)	✓	'Mon' 'Tue' 'Wed' 'Thu' 'Fri' 'Sat'
slots[].start_time	String	✓	'HH:MM' format (24hr)
slots[].end_time	String	✓	'HH:MM' format (24hr)
slots[].is_available	Boolean	✓	Can be toggled without deleting slot
effective_from	Date	✓	When this schedule takes effect
updated_at	Date	✓	Auto

6.9 DoubtSession (faculty booking records)

Field	Type	Required	Description / Constraints
_id	ObjectId	✓	Auto
student_id	ObjectId (ref: User)	✓	
faculty_id	ObjectId (ref: User)	✓	
dept	String	✓	Denormalised
subject	String	✓	Subject of the doubt
doubt_topic	String	✓	Student's description of the doubt (max 500 chars)
requested_at	Date	✓	When student submitted the booking
confirmed_at	Date	–	Set when faculty confirms

Field	Type	Required	Description / Constraints
session_datetime	Date	–	Agreed session time
meeting_link	String	–	URL from Google Meet / Zoom / Teams API
status	String (enum)	✓	'pending' 'confirmed' 'declined' 'completed'
faculty_notes	String	–	Post-session notes (max 500 chars)

6.10 ForumPost (community forum)

Field	Type	Required	Description / Constraints
_id	ObjectId	✓	Auto
author_id	ObjectId (ref: User)	✓	Student or faculty who posted
dept	String (enum)	✓	Scopes visibility — students see only their dept
subject	String	–	Optional subject tag for the post
type	String (enum)	✓	'question' 'answer' 'announcement'
content	String	✓	Post body text (max 2000 chars)
parent_id	ObjectId (ref: ForumPost)	–	Set for replies; null for top-level posts
is_grievance	Boolean	✓	Default false; true when flagged as formal grievance
grievance_ref	String	–	Auto-generated reference number e.g. 'GRV-2026-0042'
is_hidden	Boolean	✓	Default false; set true by moderator (soft delete)
created_at	Date	✓	Auto
updated_at	Date	✓	Auto

6.11 FeedbackForm (weekly student feedback)

Field	Type	Required	Description / Constraints
_id	ObjectId	✓	Auto
student_id	ObjectId (ref: User)	✓	
dept	String	✓	Denormalised — for admin filter
week	String	✓	ISO week string e.g. '2026-W35' — unique per student per week

Field	Type	Required	Description / Constraints
ratings.overall	Number (1–5)	✓	Overall platform satisfaction
ratings.content_quality	Number (1–5)	✓	Quality of study material
ratings.mcq_quality	Number (1–5)	✓	Quality of generated MCQs
feature_issue	String	–	Dropdown value: 'mcq_error' 'lockout_issue' 'forum_bug' 'other'
comments	String	–	Free-text feedback (max 500 chars)
timestamp	Date	✓	Server timestamp on submit

6.12 EmergencyContact (campus safety)

Field	Type	Required	Description / Constraints
_id	ObjectId	✓	Auto
campus	String	✓	Campus name / location identifier
contact_type	String (enum)	✓	'hospital' 'police' 'fire'
name	String	✓	Facility / service name
phone	String	✓	Primary contact number
phone_alt	String	–	Secondary contact number
address	String	✓	Full address
maps_link	String	✓	Google Maps / Apple Maps URL
updated_at	Date	✓	Auto — admin must keep current (ASM-005)

6.13 ExamTimetable (lockout source of truth)

Field	Type	Required	Description / Constraints
_id	ObjectId	✓	Auto
student_id	ObjectId (ref: User)	✓	Specific student — lockout is per-student
dept	String	✓	Denormalised
subject	String	✓	Subject being examined
exam_date	Date	✓	Date of exam

Field	Type	Required	Description / Constraints
start_time	String	✓	'HH:MM' format — lockout starts 1hr before this
end_time	String	✓	'HH:MM' format — lockout clears 1hr after this
lockout_start	Date	✓	Pre-computed: exam_date + start_time - 1hr (indexed)
lockout_end	Date	✓	Pre-computed: exam_date + end_time + 1hr (indexed)
is_manually_unlocked	Boolean	✓	Default false; admin override with audit reason
upload_batch_id	String	✓	Tracks which CSV upload created this record

7. Mailing System — Nodemailer + SMTP

All transactional emails use Nodemailer with the institution's locally hosted SMTP server. No third-party email SaaS is used in v1. All templates are HTML files with Handlebars interpolation stored in `server/src/templates/email/`.

7.1 SMTP Configuration

```
// server/src/config/mailler.js
const nodemailer = require('nodemailer');

const transporter = nodemailer.createTransport({
  host: process.env.SMTP_HOST, // Institution's local SMTP server
  port: process.env.SMTP_PORT, // 587 (STARTTLS recommended)
  secure: false, // Use STARTTLS, not SSL on port 465
  auth: {
    user: process.env.SMTP_USER,
    pass: process.env.SMTP_PASS,
  },
  tls: { minVersion: 'TLSv1.2' }, // NFR-007 compliance
});

// Verify on startup — crash fast if SMTP unreachable
transporter.verify().catch(err => {
  console.error('SMTP connection failed:', err);
  process.exit(1);
});

module.exports = transporter;
```

7.2 Email Trigger Inventory

ID	Event	Recipient	Template File	Timing	SLA
E-01	Student books doubt session	Faculty member	faculty_session_request.html	Immediate on POST /api/doubt-session	—
E-02	Faculty confirms session	Student + Faculty (both)	session_confirmed.html (includes meeting link)	Immediate on confirm	—
E-03	Faculty declines session	Student	session_declined.html	Immediate on decline	—
E-04	Unethical request flagged	HOD (dept-matched)	ethics_escalation.html	Max 5 min (OBJ-003)	5 min hard SLA via BullMQ
E-05	Hallucination report submitted	Vendor contact + IT	hallucination_report.html	Immediate	SM-006 tracks resolution
E-06	Forum grievance flagged	Caper Pvt LTD + Admin	grievance_escalation.html	Immediate + reference number	—

ID	Event	Recipient	Template File	Timing	SLA
E-07	Student distress confirmed	Campus security / counsellor	distress_alert.html	Max 2 min post confirmation	CON-005 two-step required
E-08	Weekly feedback reminder	All active students (per dept)	feedback_reminder.html	Every Monday 9:00 AM IST (cron)	feedbackCron.js

7.3 Template Requirements

- All templates in server/src/templates/email/ as .html files
- Handlebars {{ variable }} interpolation for dynamic values
- Responsive design — renders correctly on mobile + desktop email clients
- Include AI Buddy logo + institution name in header
- Plain-text fallback (.txt) for every HTML template
- PII policy: HOD/vendor emails use student_id and grievance_ref ONLY — no student name or email
- Unsubscribe footer on E-08 (feedback reminder) as required by DPDP Act 2023

7.4 Meeting Link Auto-Generation

Step	Action	Fallback
1	Faculty clicks Confirm on /faculty/session-requests	—
2	meeting.service.js calls configured provider API (Google Meet / Zoom / Teams — DEP-003)	—
3	Meeting link URL returned and stored in DoubtSession.meeting_link	—
4	E-02 email sent to both student and faculty with the link embedded	—
5	Student sees live link on /student/sessions page	—
ERR	If meeting API call fails: IT alert email sent, E-02 sent with 'link generation failed' note + IT contact	Manual coordination

8. Community Forum & Feedback Module

8.1 Forum Feature Specification

Feature	Requirement	Enforcement
Dept Scope	Students see ONLY their own dept threads	deptScope.middleware.js + MongoDB { dept } filter
Post Types	'question' 'answer' 'announcement' (HOD/Faculty only)	ForumPost.type enum + RBAC check
Moderation	HOD and Faculty can hide posts (soft delete)	is_hidden flag; hidden posts excluded from student queries
Grievance Flag	Any post flaggable as formal grievance	Triggers Grievance doc creation + E-06 email + ref number
Grievance Routing	Auto-routed to Caper Pvt LTD + Admin	E-06 email; stored in Grievance collection with status
Search	Full-text search within dept	MongoDB text index on ForumPost.content
Pagination	20 posts per page	GET /api/forum/posts?dept=&page=&limit=20
Peer Social	Likes, DMs, profiles — OUT OF SCOPE v1	Not implemented; BRD §2.1.2

8.2 Weekly Feedback Form — Full Specification

8.2.1 Trigger Flow

- feedbackCron.js fires every Monday 9:00 AM IST
- Sets feedback_due = true on all User documents where role === 'student' and is_active === true
- E-08 reminder email dispatched to all active students via Nodemailer
- On student login or forum page load: if feedback_due === true → WeeklyFeedbackModal auto-shown
- On successful submit → feedback_due = false; rate limiter prevents re-submission same week

8.2.2 Feedback Form Fields

Field	Type	UI Component	Validation	MongoDB Path
Overall Rating	Integer 1–5	Star rating (RatingInput)	Required	ratings.overall
Content Quality	Integer 1–5	Star rating	Required	ratings.content_quality
MCQ Quality	Integer 1–5	Star rating	Required	ratings.mcq_quality

Field	Type	UI Component	Validation	MongoDB Path
Feature Issue	String (enum)	Dropdown	Optional	feature_issue
Open Comments	String	Textarea	Optional, max 500 chars	comments
student_id	ObjectId	Hidden (from JWT)	Auto	student_id
week	String	Hidden (server)	Auto-computed	week
timestamp	Date	Hidden (server)	Auto-set on submit	timestamp

8.2.3 POST /api/forum/feedback — Endpoint Spec

```

POST /api/forum/feedback
Auth:    Bearer JWT (role: student only)
Rate:    Max 1 successful submission per student per ISO week (429 if exceeded)

Request Body:
{
  overall_rating:      Number (1-5, required),
  content_quality_rating: Number (1-5, required),
  mcq_quality_rating:  Number (1-5, required),
  feature_issue:       String (optional, enum),
  comments:            String (optional, max 500 chars)
}

Responses:
201 { success: true, message: 'Feedback submitted' }
400 { error: 'Validation failed', details: [ ...fieldErrors ] }
401 Unauthorized — invalid / expired JWT
403 Forbidden — role is not 'student'
429 { error: 'Already submitted this week' }
```

8.3 Admin Feedback Viewer — /admin/feedback

Feature	Specification
Access	Role: 'admin' sees all depts; role: 'hod' sees own dept only
Summary Cards	Average ratings (overall / content / MCQ) for selected week + dept — shown at top
Main Table	Columns: Week Dept Overall Content MCQ Issue Comments Timestamp
Filters	Week (date picker) · Department (dropdown) · Min Rating (slider 1–5)
Student ID	Shown in table only for 'admin' role — hidden for 'hod' (NFR-006)
CSV Export	GET /api/admin/feedback/export?week=&dept= — streams CSV download
Read-Only	No edit, delete, or reply functionality on submitted feedback

9. Exam Lockout & Timetable Integration

Requirement	Implementation Detail
Scope	Per-student, per-subject. NOT college-wide. CON-001 from BRD.
Lockout Window	1 hour before exam start_time → 1 hour after exam end_time (pre-computed in ExamTimetable.lockout_start / lockout_end)
Detection	lockoutCron.js runs every 15 min. Queries ExamTimetable where lockout_start <= NOW <= lockout_end. Sets in-memory cache (Redis optional) or DB flag.
Enforcement	lockout.middleware.js checks lockout status on every /api/llm/*, /api/student/practice/*, /api/student/mock-test* request. Returns HTTP 423 Locked with { message, unlocks_at } if active.
UI Behaviour	Student dashboard shows red 'EXAM LOCKOUT ACTIVE — unlocks at HH:MM' banner. All AI feature buttons disabled with tooltip. Emergency panel remains fully active.
Timetable Upload	Admin uploads CSV via POST /api/admin/timetable/upload. Server parses, validates, pre-computes lockout_start / lockout_end, bulk-inserts to ExamTimetable collection.
CSV Format	student_id, subject, exam_date (YYYY-MM-DD), start_time (HH:MM), end_time (HH:MM)
Manual Override	Admin can POST /api/admin/timetable/unlock/:student_id/:subject with reason. Sets is_manually_unlocked = true. AuditLog entry created. Audit trail required (compliance).
Re-enable	Auto: lockoutCron detects lockout_end has passed; clears lock. No student action needed.
Emergency Exception	lockout.middleware.js explicitly excludes /api/safety/* and /student/emergency routes.

10. Ethical Use Detection & HOD Escalation

Component	Specification
Detection Method	Every student message in /api/llm/* passes through ethics.service.js BEFORE the RAG/Gemini call. Two-layer check: (1) keyword/pattern match against EthicsConfig rules, (2) secondary Gemini prompt if pattern match inconclusive.
EthicsConfig Source	HOD-defined prohibited categories stored in EthicsConfig collection per dept. Versioned — changes logged. Default categories seeded at go-live.
Default Prohibited	Requesting direct exam answers, impersonating faculty/admin, generating fake academic documents, asking to bypass plagiarism detection, generating content for another student's assignment.
On Detection	1. Block request — return ethics violation message to student (no AI response). 2. Write EthicsFlag + AuditLog entry synchronously. 3. Enqueue HOD escalation job in BullMQ queue.
Escalation Queue	BullMQ job: dispatch E-04 email to dept HOD. Job retries on SMTP failure. Guaranteed delivery within 5 min (OBJ-003). Job completion timestamp stored in EthicsFlag.hod_notified_at.
HOD Review	HOD views flag at /hod/audit-log. Full context: student_id, request_text, category matched, severity, timestamp. Actions: 'Reviewed' 'False Positive' 'Escalate to Admin'.
False Positive	HOD marks as false_positive → EthicsFlag.hod_review_status updated. Student record not impacted. AuditLog entry appended (immutable body preserved).

11. Student Safety & Emergency Module

Component	Specification
Distress Detection	Student messages in AI Tutor (/api/llm/tutor/chat) scanned by safety.service.js in parallel (non-blocking). Gemini classifier prompt identifies distress signals (self-harm, panic, crisis language).
Two-Step Confirmation	On distress signal: AI Buddy sends in-app modal: 'It sounds like you may be struggling. Do you need help?' → Student clicks 'Yes, get help' to confirm. CON-005 compliance — no auto-escalation without confirmation.
On Confirmed Distress	E-07 email dispatched to campus security + counselling contact. In-app status panel shown to student: 'Help is on the way.' AuditLog flagged 'distress'. Timing SLA: < 2 min post-confirm.
Emergency Panel	Floating button on every screen (z-index top). Tapping opens EmergencyPanel overlay. Shows hospital / police / fire contact cards. Each card: name, phone (tap-to-call), address, Maps directions link.
24/7 Availability	Emergency panel routes explicitly excluded from lockout.middleware.js. Works when locked out. Works when session expired (minimal unauth endpoint for contact fetch).
Contact Management	Admin updates contacts at /admin/emergency-contacts. Stored in EmergencyContact collection. Live preview available before saving. Changes reflected immediately in student app.

12. Role-Based Access Control (RBAC)

All routes enforce JWT verification (auth.middleware.js) followed by role/dept checks (rbac.middleware.js). The matrix below defines access per route group. ✓ = allowed, ✗ = 403 Forbidden.

Route / Feature	Student	Faculty	HOD	Admin
/student/* (own dept only)	✓	✗	✗	✓ (all depts)
/faculty/* (own dept)	✗	✓	✗	✓
/hod/* (own dept)	✗	✗	✓	✓
/admin/*	✗	✗	✗	✓
/api/llm/* (+ lockout check)	✓	✗	✗	✓
/api/forum/posts (GET/POST)	✓ (own dept)	✓ (own dept)	✓ (own dept)	✓ (all)
/api/forum/feedback (POST)	✓	✗	✗	✗
/api/admin/feedback (GET)	✗	✗	✓ (own dept)	✓ (all)
/api/hod/audit-log (GET)	✗	✗	✓ (own dept)	✓
/api/safety/* (Emergency)	✓ (always, no lockout)	✓	✓	✓
/api/student/emergency (GET)	✓ (unauthenticated fallback)	✓	✓	✓
Cross-dept data access	✗ (deptScope middleware)	✗	✗	✓

13. Non-Functional Requirements

NFR ID	Category	Requirement	Target / Threshold
NFR-001	Scalability	Support concurrent users during peak exam-prep	3,000+ concurrent — horizontal scaling on AWS/Azure
NFR-002	Availability	System uptime outside per-student exam lockout windows	≥ 99.5% — monitored weekly (SM-007)
NFR-003	Performance	AI response delivery at P95 under peak load	≤ 3 seconds — Groq fallback helps during Gemini rate-limit
NFR-004	Data Retention	Flagged + hallucination report data retention period	Per institutional data-retention policy
NFR-005	Platform	Accessible on mobile (iOS/Android) and web (desktop)	Responsive React web app; PWA for mobile in v1
NFR-006	Security	Zero cross-role data leakage between all four roles	Zero unauthorized access — enforced by RBAC + deptScope middleware
NFR-007	Encryption	Encrypt data in transit and at rest	TLS 1.2+ (Nodemailer + HTTPS) / AES-256 (MongoDB Atlas at rest)
NFR-008	Usability	Multi-language support (open item)	TBD — pending student demographics survey
NFR-009	Safety Availability	Emergency features available 24/7 independent of exam session	24/7 — excluded from lockout middleware
NFR-010	Auditability	Audit logs immutable and searchable by compliance reviewers	Immutable (no delete/update on AuditLog) + role-restricted read

14. API Endpoint Register

All REST endpoints. Base URL: <https://api.aibuddy.institution.edu/api>

14.1 Auth

Method	Endpoint	Role	Description
POST	/auth/login	All	SSO OAuth 2.0 callback → JWT + refresh token
POST	/auth/logout	All	Invalidate session + blacklist JWT
POST	/auth/refresh	All	Refresh expired JWT using refresh token

14.2 LLM / MCQ / Learning

Method	Endpoint	Role	Description
POST	/llm/mcq/generate	Student	RAG + Gemini → MCQ set for subject+topic. Body: {subject, topic, count}
POST	/llm/mock-test/generate	Student	Generate full mock test. Body: {subject, duration_minutes}. Returns MockTest_id
POST	/llm/mock-test/:id/submit	Student	Submit answers → score + topic breakdown computed
POST	/llm/learning-path/generate	Student	Generate / refresh personalised learning path from student persona
POST	/llm/tutor/chat	Student	RAG-powered AI tutor. Body: {message}. Returns {response, rag_sources[]}
GET	/student/question-bank	Student	Query past paper bank. Params: dept, subject, topic, year, page

14.3 Faculty & Sessions

Method	Endpoint	Role	Description
GET	/faculty/availability	Student / Faculty	Get faculty availability slots. Params: faculty_id, date_range
POST	/faculty/availability	Faculty	Set/update availability windows. Body: {slots[]}
POST	/doubt-session	Student	Book session. Body: {faculty_id, subject, doubt_topic}. Triggers E-01
POST	/doubt-session/:id/confirm	Faculty	Confirm session → generates meeting link → sends E-02

Method	Endpoint	Role	Description
POST	/doubt-session/:id/decline	Faculty	Decline session → sends E-03 to student
GET	/doubt-session/my	Student / Faculty	List own sessions with status + meeting links

14.4 Forum & Feedback

Method	Endpoint	Role	Description
GET	/forum/posts	Student/Faculty/HOD	List dept-scoped posts. Params: page, limit, subject, type
POST	/forum/posts	Student/Faculty/HOD	Create post. Body: {type, content, subject?}
POST	/forum/posts/:id/reply	Student/Faculty/HOD	Reply to post. Body: {content}
POST	/forum/posts/:id/flag-grievance	Student	Flag post as grievance → creates Grievance doc + E-06 email
DELETE	/forum/posts/:id	HOD / Faculty	Soft-delete (hide) post. Sets is_hidden=true
POST	/forum/feedback	Student	Submit weekly feedback form. Max 1 per student per week.
GET	/admin/feedback	HOD / Admin	View all feedback submissions. Params: week, dept, min_rating
GET	/admin/feedback/export	Admin	Download feedback as CSV stream

14.5 Admin, Lockout & Safety

Method	Endpoint	Role	Description
POST	/admin/timetable/upload	Admin	Upload exam schedule CSV → parse + populate ExamTimetable
POST	/admin/timetable/unlock/:sid/:subject	Admin	Manual lockout override. Body: {reason}. Creates AuditLog entry.
GET	/lockout/status	Student	Check own current lockout state. Returns {is_locked, unlocks_at?}
GET	/admin/emergency-contacts	All	Fetch emergency contacts for campus
PUT	/admin/emergency-contacts/:id	Admin	Update emergency contact details
POST	/safety/distress	Student	Submit confirmed distress signal → E-07 email dispatched
POST	/hallucination/report	Student	Submit hallucination report → E-05 email + AuditLog flag
GET	/hod/audit-log	HOD / Admin	Paginated immutable audit log. Params: date_range, flag_type, status
PUT	/hod/ethics-config	HOD	Update dept prohibited request categories

Method	Endpoint	Role	Description
GET	/admin/dashboard	Admin	System-wide KPI aggregates for dashboard charts
POST	/admin/users	Admin	Create user. Body: {institution_id, name, email, role, dept, ...}
PUT	/admin/users/:id	Admin	Update user role/dept/semester/active status

15. Glossary

Term	Definition
Gemini 2.0 Flash	Google's free-tier LLM with 1M token context window, native JSON mode, and text-embedding-004 embedding model. Primary LLM for AI Buddy.
Groq	Free LLM inference platform (Llama 3.3 70B model). Used as automatic fallback when Gemini API returns 429 or 5xx.
RAG	Retrieval-Augmented Generation — AI architecture that grounds LLM outputs in a specific knowledge base (institution documents) to reduce hallucination.
text-embedding-004	Google's embedding model (free with Gemini API key). Produces 768-dim vectors used for vector similarity search in MongoDB Atlas.
Atlas Vector Search	MongoDB Atlas feature enabling semantic similarity search on embedded vectors. Available on free M0 tier.
BullMQ	Node.js job queue library used to guarantee 5-minute HOD escalation SLA (OBJ-003) with retry on SMTP failure.
MCQ	Multiple Choice Question — generated by Gemini 2.0 Flash from RAG-retrieved subject content per the schema in Section 5.5.
Lockout	Per-student, per-subject suspension of all AI features from 1hr before to 1hr after each exam. Enforced by <code>lockout.middleware.js</code> .
Persona	Student profile: <code>dept</code> , <code>semester</code> , <code>enrolled_subjects[]</code> , <code>weak_topics[]</code> , <code>mock_scores[]</code> , <code>exam_schedule[]</code> . Input to learning path generation.
RBAC	Role-Based Access Control — enforced by <code>rbac.middleware.js</code> + <code>deptScope.middleware.js</code> on every API route.
DPDP Act 2023	Digital Personal Data Protection Act 2023 — India's primary data privacy legislation. All student PII handling must comply.
Hallucination	AI output presenting factually incorrect or fabricated information as accurate. Reported via <code>/api/hallucination/report</code> .
Nodemailer	Node.js library used to send all 8 transactional email types via the institution's local SMTP server.
EthicsConfig	Per-dept HOD-defined rules listing prohibited request categories. Versioned MongoDB collection; input to <code>ethics.service.js</code> .
Grievance	A formally flagged forum post. Creates a Grievance document with a unique reference number; routed to Caper Pvt LTD + Admin.
CON-005	BRD constraint: distress escalation requires two-step confirmation before automated alert is sent — not triggered on first signal.
Caper Pvt LTD	Student project team: Om Pandey (Scrum Master), Shivam Mishra (PM), Rehan Mulani (BA), Hitarth Patel (Solution Architect & TL).

— *End of Software Requirements Document v2.0* —

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